

Netflix Content Analysis.

Introduction:

This project is designed to showcase my expertise in Exploratory Data Analysis (EDA) by cleaning, modelling, and analyzing data to uncover key insights and KPIs. It provides a comprehensive understanding of both current and historical trends, as well as the overall characteristics of the data. Additionally, I've included some predictive analysis to address key questions that could influence performance and trends moving forward. The goal of the dashboard is to present these insights in a simple and interactive way, making it easier to communicate the most important findings to stakeholders. The data for this project was sourced from Kaggle.com, and the technologies used include Excel, Power Query, and Power BI.

The dataset consists of the following parameters:

1. show_id: Unique identifier for the movie or TV show.
2. type: Specifies whether it is a movie or a TV show.
3. title: Name of the program.
4. director: Director responsible for the production, applicable only to movies.
5. cast: Main cast members.
6. date_added: Date when the program was added to the platform.
7. release_year: Year the program was released.
8. rating: Age-based content classification.
9. duration: Number of seasons (for TV shows) or runtime in minutes (for movies).
10. listed_in: Genre classification of the program.
11. description: A brief summary of the program.

Objective:

The main goal of this project is to develop a customized dashboard tailored specifically to the client's needs, with a focus on presenting important insights in a clear and accessible way. To achieve this, I have used exploratory data analysis (EDA) to explore and understand the data, data modelling techniques to analyze trends and patterns, and effective data presentation methods to ensure the insights are easily digestible. Additionally, I've made sure that the design of the dashboard aligns with the company's branding, particularly by incorporating their preferred colour palette to ensure a cohesive and professional look. The aim is to make the dashboard not only functional and informative but also visually appealing and consistent with the company's identity.

These are some of the objectives that I aim to derive.

1. Genres and its distribution
2. Top countries producing content across the platform
3. Popular type of content produced/consumed across various countries.
4. Ratings of the movies.
5. Year wise content addition trend.
6. Latest release date for a particular genre, country or rating.

Plan of action:

- The initial step in the project is to import the dataset and profile it within the working environment. This process allows me to understand the structure of the data, identify any missing or inconsistent values, and get a sense of the overall quality of the dataset. By performing this initial profiling, I can ensure that the data is properly prepared for the subsequent stages of cleaning, manipulation, and analysis, setting the foundation for accurate and reliable insight analysis.

show_id	type	title	director	cast	country
s1	Movie	Dick Johnson Is Dead	Kirsten Johnson		United States
s2	TV Show	Blood & Water		Ama Omatola, Khosi Ngema, Gail Mablane, Thabang Molaba, Dillon W...	South Africa
s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabila Akkari, Sofia Lesaffr...	
s4	TV Show	Jailbirds New Orleans			
s5	TV Show	Kota Factory		Mayur More, Jitendra Kumar, Ranjan Raj, Alan Khan, Ahsaas Channa, ...	India
s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gifford, Hamish Linklater, Henry Thomas, Kristin Leh...	
s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, Sofia Carson, Liza Ko...	
s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyallumika Ogunlanso, Alexandra Duah, Nick Medley, ...	United States, Ghana, Bur
s9	TV Show	The Great British Baking Show	Andy Devonshire	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Hollywood	United Kingdom
s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, Timothy Olyphant, Dave...	United States
s11	TV Show	Vendetta: Truth, Lies and The Mafia			
s12	TV Show	Bangkok Breaking	Kongkiat Komesiri	Sukollawat Kanarot, Sushar Manaying, Pavarit Mongkolpisit, Sahajak B...	
s13	Movie	Je Suis Karl	Christian Schwowchow	Luna Wedler, Jannis Niewöhner, Milan Peschel, Edin Hasanović, Anna ...	Germany, Czech Republic
s14	Movie	Confessions of an Invisible Girl	Bruno Garotti	Klara Castanho, Lucca Picon, Júlia Gomes, Marcus Bessa, Kiria Malheir...	
s15	TV Show	Crime Stories: India Detectives			
s16	TV Show	Dear White People		Logan Browning, Brandon P. Bell, Defton Horton, Antoinette Robertson...	United States
s17	Movie	Europe's Most Dangerous Man: Otto Skorzeny in Spain	Pedro de Echave García, Pablo Azorín Williams		
s18	TV Show	Falsa identidad		Luis Ernesto Franco, Camila Sodi, Sergio Goyri, Samadhi Zendejas, Eda...	Mexico
s19	Movie	Intrusion	Adam Salky	Freida Pinto, Logan Marshall-Green, Robert John Burke, Megan Elisabe...	
s20	TV Show	Jaguar		Blanca Suárez, Iván Marcos, Óscar Casas, Adrián Lastra, Francesc Garri...	
s21	TV Show	Monsters Inside: The 24 Faces of Billy Milligan	Olivier Megaton		
s22	TV Show	Resurrection: Ertugrul		Engin Altan Düzgün, Serdar Gökhan, Hülya Darcan, Kaan Tuganer, Ert...	Turkey
s23	Movie	Avsar Shammugh	K.S. Ravikumar	Kamul Hassan, Meena, Gemini Ganesan, Heera Rajgopal, Nassar, S.P. B...	
s24	Movie	Go! Go! Cory Carson: Christy Takes the Wheel	Alex Woo, Stanley Moore	Masie Benson, Paul Kiliam, Kerry Gudjohnsen, AC Lim	
s25	Movie	Jeans	S. Shankar	Prashanth, Aishwarya Rai Bachchan, Sri Lakshmi, Nassar	India
s26	TV Show	Love on the Spectrum		Brooke Satchwell	Australia
s27	Movie	Minsara Kanavu	Rajiv Menon	Arvind Swamy, Kajol, Prabhu Deva, Nassar, S.P. Balasubrahmanyam, GL...	

- The raw data has some issues, such as irrelevant columns, blank cells, and duplicates. The next step is to clean and organize the data. This involves removing unnecessary columns, filling in or removing blank cells, and getting rid of any duplicate entries to make the data more accurate and ready for analysing.

director
Kirsten Johnson
Unknown Director
Julien Leclercq
Unknown Director
Unknown Director
Mike Flanagan
Robert Cullen, José Luis Ucha
Haile Gerima
Andy Devonshire
Theodore Melfi
Unknown Director
Kongkiat Komesiri
Christian Schwowchow
Bruno Garotti
Unknown Director
Unknown Director
Pedro de Echave García, Pablo Azorín Williams
Unknown Director
Adam Salky
Unknown Director
Olivier Megaton
Unknown Director
K.S. Ravikumar
Alex Woo, Stanley Moore
S. Shankar
Unknown Director
Rajiv Menon

- We notice that the director column has many missing names. There are two ways to handle this: we could either leave the blank spaces as they are or fill them with a placeholder like "unknown director." In this case, we chose to replace the blank spaces with "unknown director" to make it easier to analyze the data later on.

country
United States
South Africa
United States
United States
India
United States
United States
United States, Ghana, Burkina Faso, U...
United Kingdom
United States
United States
United States
Germany, Czech Republic
United States
United States
United States
United States
Mexico
United States
United States
United States
Turkey
United States
United States
India
Australia
United States

- The country column also has some blank cells. We have two options: we can either leave them blank or fill them with an estimated value. In this case, we decided to fill the blank cells with "United States," since it produces the majority of the content in the dataset.
- We also made adjustments to the Date column by changing the "locale type" to make sure the dates are consistent across the entire dataset. This helps avoid any confusion caused by different date formats.

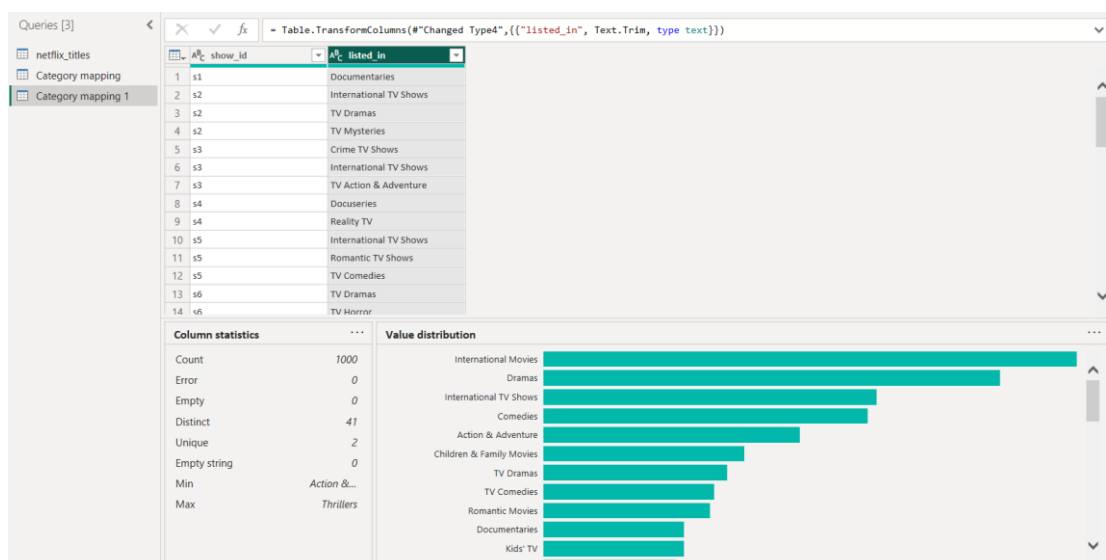
duration	listed_in	description	Text Before Delimiter	Duration in mins
90 min	Documentaries	As her father nears the end of his life, filmmaker Kirsten Johnson stage...		90 min
2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town teen sets out to prove wh...	Ama Qamata, Khosi Ngema, Gail Mabalane	null
1 Season	Crime TV Shows, International TV Shows, TV Action & Adventure	To protect his family from a powerful drug lord, skilled thief Mehdi an...	Sami Bouajila, Tracy Gotoas, Samuel Jouy	null
1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down among the incarcerated wom...		null
2 Seasons	International TV Shows, Romantic TV Shows, TV Comedies	In a city of coaching centers known to train India's finest collegiate mi...	Mayur More, Jitendra Kumar, Ranjan Raj	null
1 Season	TV Dramas, TV Horror, TV Mysteries	The arrival of a charismatic young priest brings glorious miracles, omin...	Kate Siegel, Zach Gilford, Hamish Linklater	null
91 min	Children & Family Movies	Equestria's divided. But a bright-eyed hero believes Earth Ponies, Pega...	Vanessa Hudgens, Kimiko Glenn, James Marsden	91 min
125 min	Dramas, Independent Movies, International Movies	On a photo shoot in Ghana, an American model slips back in time, bec...	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra Duah	125 min
9 Seasons	British TV Shows, Reality TV	A talented batch of amateur bakers face off in a 10-week competition,...	Mel Giedroyc, Sue Perkins, Mary Berry	null
104 min	Comedies, Dramas	A woman adjusting to life after a loss contends with a feisty bird that'...	Melissa McCarthy, Chris O'Dowd, Kevin Kline	104 min
1 Season	Crime TV Shows, Docuseries, International TV Shows	Sicily boasts a bold "Anti-Mafia" coalition. But what happens when tho...		null
1 Season	Crime TV Shows, International TV Shows, TV Action & Adventure	Struggling to earn a living in Bangkok, a man joins an emergency rescu...	Sukollawat Kanarot, Sushar Manaying, Pavarit Mongkolpisit	null
127 min	Dramas, International Movies	After most of her family is murdered in a terrorist bombing, a young w...	Luna Wedler, Jannis Niewöhner, Milan Peschel	127 min
91 min	Children & Family Movies, Comedies	When the clever but socially-awkward Teté joins a new school, she'll d...	Klara Castanho, Lucca Picon, Júlia Gomes	91 min
1 Season	British TV Shows, Crime TV Shows, Docuseries	Cameras following Bengaluru police on the job offer a rare glimpse int...		null
4 Seasons	TV Comedies, TV Dramas	Students of color navigate the daily slights and slippery politics of life a...	Logan Browning, Brandon P. Bell, DeRon Horton	null
67 min	Documentaries, International Movies	Declassified documents reveal the post-WWII life of Otto Skorzeny, a c...		67 min
2 Seasons	Crime TV Shows, Spanish-Language TV Shows, TV Dramas	Strangers Diego and Isabel flee their home in Mexico and pretend to b...	Luis Ernesto Franco, Camila Sodi, Sergio Goyri	null
94 min	Thrillers	After a deadly home invasion at a couple's new dream house, the trau...	Freida Pinto, Logan Marshall-Green, Robert John Burke	94 min
1 Season	International TV Shows, Spanish-Language TV Shows, TV Action & Adv...	In the 1960s, a Holocaust survivor joins a group of self-trained spies w...	Blanca Suárez, Iván Marcos, Óscar Casas	null
1 Season	Crime TV Shows, Docuseries, International TV Shows	In the late 1970s, an accused serial rapist claims multiple personalities ...		null
5 Seasons	International TV Shows, TV Action & Adventure, TV Dramas	When a good deed unwittingly endangers his clan, a 13th-century Tur...	Engin Altan Düzgatan, Serdar Gökhan, Hülya Darcan	null
161 min	Comedies, International Movies	Newly divorced and denied visitation rights with his daughter, a dotin...	Kamal Hassan, Meena, Gemini Ganesan	161 min
61 min	Children & Family Movies	From arcade games to sled days and hiccup cures, Cory Carson's curio...	Maisie Benson, Paul Killam, Kerry Gudjohnsen	61 min
166 min	Comedies, International Movies, Romantic Movies	When the father of the man she loves insists that his twin sons marry t...	Prashanth, Aishwarya Rai Bachchan, Sri Lakhshmi	166 min
2 Seasons	Docuseries, International TV Shows, Reality TV	Finding love can be hard for anyone. For young adults on the autism s...	Brooke Satchwell	null
147 min	Comedies, International Movies, Music & Musicals	A tangled love triangle ensues when a man falls for a woman studying ...	Arvind Swamy, Kajol, Prabhu Deva	147 min

- We noticed that the Duration column has two different types of data: one shows the duration in minutes (for movies), and the other shows the number of seasons (for TV shows). To make this data easier to work with, we created a new column. In this new column, the duration for movies is listed in minutes, while the duration for TV shows is marked as "null." This extra column will help when creating the dashboard or analyzing key insights related to the duration of movies.
- We also changed the text in the description and director fields to lowercase. This makes it easier to analyze the data and apply other techniques during the exploration and analysis process.

Table.AddColumn("#Lowercased Text", "Murder mystery?", each if Text.Contains([description], "murder") then "yes" else if Text.Contains([description], "kill") then "yes" else if Text.Contains([description], "death") then "yes" else if Text.Contains([description], "assassinate") then "yes" else "no")

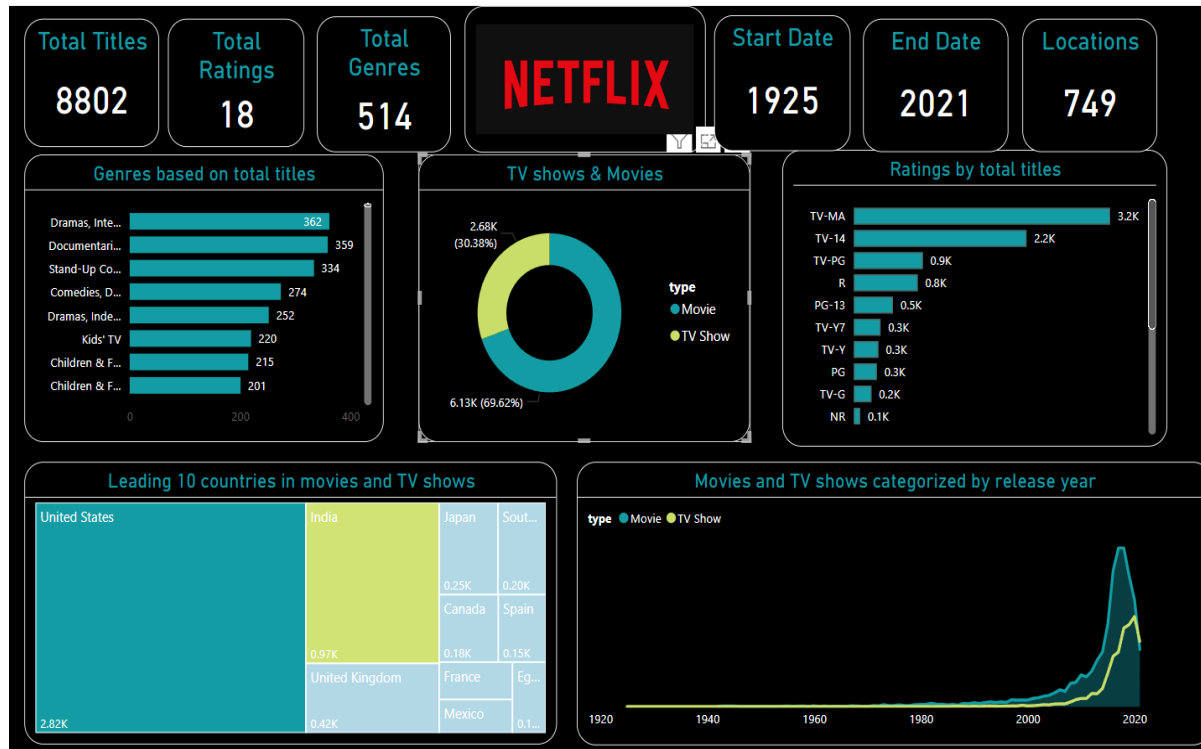
	Text Before Delimiter	Duration in mins	First category	Murder mystery?
1	a end of his life, filmmaker kirsten johnson stage...	90	Documentaries	yes
2	a party, a cape town teen sets out to prove whe...	null	International TV Shows	no
3	om a powerful drug lord, skilled thief mehdi and...	null	Crime TV Shows	yes
4	oilet talk go down among the incarcerated wom...	null	Docuseries	no
5	enters known to train india's finest collegiate m...	null	International TV Shows	no
6	hatic young priest brings glorious miracles, omin...	null	TV Dramas	no
7	it a bright-eyed hero believes earth ponies, pega...	91	Children & Family Movies	no
8	hana, an american model slips back in time, beco...	125	Dramas	no
9	hateur bakers face off in a 10-week competition, ...	null	British TV Shows	no
10	life after a loss contends with a feisty bird that's...	104	Comedies	no
11	iti-mafia" coalition. but what happens when tho...	null	Crime TV Shows	no
12	ing in bangkok, a man joins an emergency rescu...	null	Crime TV Shows	no
13	ly is murdered in a terrorist bombing, a young w...	127	Dramas	yes
14	ocially awkward teté joins a new school, she'll d...	91	Children & Family Movies	no
15	hgaluru police on the job offer a rare glimpse int...	null	British TV Shows	no
16	gate the daily slights and slippery politics of life a...	null	TV Comedies	no
17	ts reveal the post-wwii life of otto skorzeny, a clo...	67	Documentaries	no
18	abel flee their home in mexico and pretend to b...	null	Crime TV Shows	no
19	hvasion at a couple's new dream house, the trau...	94	Thrillers	no
20	ust survivor joins a group of self-trained spies w...	null	International TV Shows	no
21	ccused serial rapist claims multiple personalities ...	null	Crime TV Shows	no
22	wittingly endangers his clan, a 13th-century turki...	null	International TV Shows	no
23	anied visitation rights with his daughter, a dotin...	161	Comedies	no
24	clad down and hison eurus, eurus eurus's eurus...	61	Children & Family Movies	no

- We added a new column to check if a movie belongs to the "Murder mystery" genre. We set the condition to include movies that have keywords like "kill," "death," or "assassinate" in the description. This helps identify which movies fit that genre.
- We know that the quality of television programs and movies, as well as viewers' preferences, have changed a lot in recent years. So, to focus on more recent trends, we will filter the dataset to include only records with a release year from 2000 and beyond.
- We have also excluded or hidden columns like the description to keep the data more relevant. Having clear and concise data helps the dashboard run smoothly and ensures that only the most important information is displayed.



- Additionally, since each title is linked to multiple genres, we created a separate query called "Category mapping 1." This query splits the 'listen-in' column to show each individual genre along with its corresponding 'show ID.' This step helped us gain important insights, such as the total number of genres and how the titles are distributed across different genres.

Key Findings & insights:



Key Takeaways:

- **Movies** make up **68%** of the total content, while **TV shows** make up **32%**.
- The platform offers content across **42 genres**.
- A total of **8278 titles** have been released **since 2000**.
- The **peak in content** release occurred **during 2017-2018**.
- The top 3 genres in both movies and TV shows are **International, Dramas, and Comedies**. International includes a mix of various genres.
- Most content is rated **TV-MA** or **TV-14**, with **TV-MA** being the most common rating.
- The countries producing the most movies are:
 1. **United States** (over 40% share)
 2. **India** (10%)
 3. **UK** (about 5%)
 4. **Japan and South Korea**.
- **Japan and South Korea** produce more **TV shows** than movies, especially in **International TV** and **Anime** genres.
- There was a sharp rise in movies and TV shows from **2015 to 2018**, with **2018** seeing the highest number produced. However, there was a decline in production after **2020**, likely due to the **Covid-19 pandemic**.

Conclusion:

This Netflix Content Analysis shows how data-driven insights can reveal key trends in content, audience preferences, and platform performance. By cleaning and organizing data using Extract, transform and load processes, we gained valuable insights into Netflix's strategy, such as the dominance of movies (68%) over TV shows (32%) and the leading role of countries like the U.S., India, and the UK in content production. Interestingly, Japan and South Korea produce more TV shows, especially in Anime and International genres. We also found that most content is aimed at mature audiences (TV-MA and TV-14). The analysis highlights the growth of content from 2015 to 2018, followed by a decline after 2020 due to the pandemic. Using data modelling and keyword classification, we also identified popular genres like murder mysteries. The interactive dashboard makes these insights easy to explore, and future analysis on viewer engagement and regional performance could further optimize Netflix's content strategy.

Drive link:

<https://drive.google.com/file/d/10D0R1joeZvepwff45DcjMA6lrpcDQMTj/view?usp=sharing>